SP_000
SP_001
SP_002
SP_003
SP_004
SP_005
SP_006
SP_007
SP_008
SP_009

101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

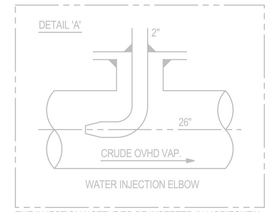
(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

Vessel
Filter
Pump / Compressor
Unknown
Tank
Tank
Vessel
Heat Exchanger
Vessel
Tank
Vessel
Temperature Indicator
Pressure Indicator
Temperature Indicator
Flow Transmitter
Flow Transmitter
Flow Primary Element
Pressure Valve
Flow Transmitter
Flow Primary Element
Analyzer Transmitter
Valve (generic)

101-E-102A/B/C/D

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
4		ISSUED FOR CONSTRUCTION		10.03.22	SC	MW	VD	PV
3		ISSUED FOR CONSTRUCTION		30.09.21	SC	MW	VD	PV
2		ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION		02.09.20	RMB	KG	RS	SG
1		ISSUED FOR ENGINEERING/DESIGN		11.08.20	JO	KG	RS	SG
0		ISSUED FOR REVIEW		13.03.20	LC	KG	RS	SG

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
1	14.12.18	REVISED AS MARKED		PP	UA/AT	GN	AM	
0	15.11.18	ISSUED FOR PACKAGE		S.K.L	UA/AT	GN	AM	

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

PMC: ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

OWNER: एच पी सी एल राजस्थान रिफाइनरी लिमिटेड
HPCL RAJASTHAN REFINERY LTD
RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमेन्टेशन डायग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

अनुमाप SCALE	कार्य संख्या JOB NO.	विभाग DEPT.	अनुभाग SECTN.	इकाई UNIT	आरेख संख्या DWG. No.	संशो REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. B224-CDU-TPL-101-1112 — पी एल आरेख संख्या — संशो REV. 4

DOCUMENT REVIEW DETAILS
1 No Comments. Doc. B224-CDU-TPL-101-1112 Rev. 4
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Returned for Records. V Void.
Signature with Date 24-AUG-2022
Name Reviewer UMER AFZAL (C889)
Page No. 1 of 2

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